Formal Analysis of the Apriori Induction Engine in TR-Core

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Abstract

We present a formal complexity analysis of the Apriori-style induction engine implemented in `BehavioralProgramV3` (TR-Core Phase 3). The engine discovers compositional rules as conjunctions of literals over a finite attribute domain, using coverage-based pruning and a binomial statistical filter. We derive worst-case and expected-case bounds for time and space complexity, prove correctness of the anti-monotonicity pruning, and characterize the statistical guarantees of the binomial filter.

1 Problem Setup and Notation

1.1 Attribute Domain

Let $\mathcal{A} = \{A_1, \ldots, A_m\}$ be a set of m categorical attributes. Each attribute A_i has a finite domain $\mathrm{dom}(A_i)$ of size $d_i = |\mathrm{dom}(A_i)|$. A *literal* is a pair (A_i, v) with $v \in \mathrm{dom}(A_i)$, denoted $A_i = v$. The set of all possible literals is

$$\mathcal{L} = \bigcup_{i=1}^{m} \{A_i\} \times \mathrm{dom}(A_i), \qquad |\mathcal{L}| = \sum_{i=1}^{m} d_i =: L.$$

In TR-Core v3/v4: $m = 5$ or 8, $d_i \in \{2, 3, 4\}$, so $L \in [15, 25]$.

1.2 Conjunctions and Arity

A *conjunction* (candidate condition) of arity k is a set $C \subseteq \mathcal{L}$ with $|C| = k$ and at most one literal per attribute. The set of all valid conjunctions of arity $\leq K$ is

$$\mathcal{C}_{\leq K} = \bigcup_{k=1}^{K} \{C \subseteq \mathcal{L} : |C| = k,\ \forall A_i, |C \cap (\{A_i\} \times \mathrm{dom}(A_i))| \leq 1\}.$$

Total candidates:

$$|\mathcal{C}_{\leq K}| = \sum_{k=1}^{K} \sum_{1 \leq i_1 < \cdots < i_k \leq m} \prod_{j=1}^{k} d_{i_j}.$$

For uniform domains $d_i = d$, this is $\sum_{k=1}^{K} \binom{m}{k} d^k \leq (1+d)^m - 1$.

1.3 Data and Coverage

The agent collects *explore trials* $\mathcal{D} = \{(x^{(t)}, a^{(t)}, y^{(t)})\}_{t=1}^{N}$ where $x^{(t)} \in \mathcal{X} = \prod_i \mathrm{dom}(A_i)$ is a context, $a^{(t)} \in \mathcal{Y}$ is an action chosen uniformly at random (exploration), and $y^{(t)} \in \{0, 1\}$ is success. For a fixed action a, define the filtered dataset

$$\mathcal{D}_a = \{(x, y) \in \mathcal{D} : a^{(t)} = a\}, \qquad N_a = |\mathcal{D}_a|.$$

For a conjunction C, its *coverage* and *hits* on $\mathcal{D}_a$ are

$$\mathrm{cov}_a(C) = |\{(x, y) \in \mathcal{D}_a : x \models C\}|, \qquad \mathrm{hits}_a(C) = |\{(x, y) \in \mathcal{D}_a : x \models C \wedge y = 1\}|.$$

Precision is $\mathrm{prec}_a(C) = \mathrm{hits}_a(C)/\mathrm{cov}_a(C)$ (undefined if $\mathrm{cov}_a(C) = 0$).

1.4 Induction Criteria

A candidate C for action a is *materialized* as a rule iff:

1. $\mathrm{cov}_a(C) \geq c_{\min}$ (coverage floor)
2. $\mathrm{hits}_a(C) \geq h_{\min}$ (minimum successes)
3. $\mathrm{prec}_a(C) \geq \theta$ (precision threshold)
4. p-value $= \mathbb{P}_{\mathrm{Binom}(N_a, p_a)}(\mathrm{hits} \geq \mathrm{hits}_a(C)) \leq \alpha$ (binomial filter)

where $p_a = \mathrm{hits}_a(\emptyset)/N_a$ is the base success rate for action a. Default parameters: $c_{\min} = 4$, $h_{\min} = 5$, $\theta = 0.95$, $\alpha = 10^{-3}$, $K = 4$.

2 Algorithm: Apriori Induction with Binomial Filter

Algorithm 1: Apriori Induction for One Action

Input: $\mathcal{D}_a$, L literals, K, thresholds $(c_{\min}, h_{\min}, \theta, \alpha)$ $S_1 \leftarrow \{\{l\} : l \in \mathcal{L}\}$ **for** $k = 1$ to K $S_k^{\text{supported}} \leftarrow \emptyset$ **for** $C \in S_k$ Compute $\mathrm{cov}_a(C), \mathrm{hits}_a(C)$ via bitmask intersection **if** $\mathrm{cov}_a(C) \geq c_{\min}$ $S_k^{\text{supported}} \leftarrow S_k^{\text{supported}} \cup \{C\}$ **if** $\mathrm{hits}_a(C) \geq h_{\min} \wedge \mathrm{prec}_a(C) \geq \theta \wedge p\text{-value}(C) \leq \alpha$ Materialize rule $C \to a$ **end if end if end for if** $S_k^{\text{supported}} = \emptyset$ **break end if** $S_{k+1} \leftarrow \mathrm{GenCandidates}(S_k^{\text{supported}})$ **end for**

2.1 Candidate Generation (Apriori Join)

GenCandidates(S) produces all $(k+1)$-literal sets whose every k-subset is in S:

$$\mathrm{GenCandidates}(S) = \{C \cup \{l\} : C \in S,\ l \in \mathcal{L} \setminus C,\ \forall C' \subset C \cup \{l\}, |C'| = k \implies C' \in S\}.$$

In the implementation (line 154-169 of `program_v3.py`), this is done by extending each supported C with literals not in C, then sorting for determinism.

3 Complexity Analysis

3.1 Worst-Case Time Complexity

Theorem 1 (Worst-Case Candidate Enumeration)**.** *The worst-case number of candidates evaluated across all arities* $1 \dots K$ *is*

$$T_{cand} = \sum_{k=1}^{K} \binom{L}{k} = \mathcal{O}(L^K).$$

Proof. Without pruning, every subset of $\mathcal{L}$ of size $\leq K$ is a candidate. The sum of binomial coefficients is bounded by $(eL/K)^K$ for $K \ll L$, i.e., $\mathcal{O}(L^K)$. □

Theorem 2 (Worst-Case Coverage Computation)**.** *For a single candidate* C*, computing* $\mathrm{cov}_a(C)$ *and* $\mathrm{hits}_a(C)$ *over* N_a *trials takes* $\mathcal{O}(N_a)$ *time naively, or* $\mathcal{O}(1)$ *with precomputed bitmasks (as implemented). With bitmasks:*

$$T_{eval} = \mathcal{O}(N_a \cdot L) \text{ preprocessing} + \mathcal{O}(T_{cand}) \text{ evaluation.}$$

Proof. Preprocessing: tokenize each trial into a bitmask of length L ($\mathcal{O}(N_a L)$). Evaluation: bitwise AND and popcount are $\mathcal{O}(1)$ on 32/64-bit words (here $L \leq 32$ fits in `uint32`). □

Corollary 1 (Overall Worst-Case Time)**.**

$$T_{worst} = \mathcal{O}(N_a L + L^K).$$

For TR-Core: $N_a \sim 2500$*,* $L \sim 20$*,* $K = 4 \implies L^K = 160{,}000$ *candidates, feasible.*

3.2 Expected Complexity Under Anti-Monotonicity

Definition 1 (Anti-Monotonicity of Coverage)**.** *Coverage is anti-monotone: if* $C \subseteq C'$ *then* $\mathrm{cov}_a(C') \leq \mathrm{cov}_a(C)$*.*

Lemma 1 (Apriori Pruning Correctness)**.** *If* $\mathrm{cov}_a(C) < c_{\min}$*, then for all* $C' \supset C$*,* $\mathrm{cov}_a(C') < c_{\min}$*.*

Proof. Direct from anti-monotonicity: $C \subseteq C' \implies \mathrm{cov}_a(C') \leq \mathrm{cov}_a(C) < c_{\min}$. □

Theorem 3 (Expected Candidates Under Pruning)**.** *Let* $p_k = \mathbb{P}(\mathrm{cov}_a(C) \geq c_{\min})$ *for a random* k*-literal conjunction. Assume* p_k *decays exponentially:* $p_k \leq \beta^k$ *for some* $\beta < 1$*. Then expected candidates:*

$$\mathbb{E}[T_{cand}] = \sum_{k=1}^{K} \binom{L}{k} p_{k-1} \leq \sum_{k=1}^{K} \binom{L}{k} \beta^{k-1} = \mathcal{O}((1+\beta L)^K).$$

Proof. A k-candidate is generated only if its $(k-1)$-subsets are supported. By union bound, probability a random k-candidate is generated $\leq \beta^{k-1}$. Sum over all $\binom{L}{k}$ candidates. □

Remark 1. *In practice,* $c_{\min} = 4$ *and sparse data (*$N_a \sim 600$*) make* p_k *drop sharply. Phase 3 logs show* ~ 100 *rules materialized per seed from* $\sim 10^5$ *initial candidates, implying effective branching factor* $\ll L$*.*

3.3 Space Complexity

Theorem 4 (Space Complexity)**.**

$$S = \mathcal{O}(N_a + L + R \cdot K)$$

where R is the number of materialized rules. Bitmasks: $\mathcal{O}(N_a)$; literal index: $\mathcal{O}(L)$; rules: $\mathcal{O}(RK)$.

In TR-Core: $R \sim 100$, $K = 4$, $N_a \sim 2500$ $\implies$ negligible memory.

4 Statistical Guarantees of the Binomial Filter

4.1 Filter Definition

For candidate C with $\text{cov}_a(C) = n$, $\text{hits}_a(C) = h$, the filter computes

$$p\text{-value}(C) = \sum_{j=h}^{n} \binom{n}{j} p_a^j (1 - p_a)^{n-j} = \mathbb{P}_{X \sim \text{Binom}(n, p_a)}(X \geq h).$$

Accept if p-value $\leq \alpha$.

4.2 Type I Error (False Positive)

Theorem 5 (Per-Candidate Type I Error)**.** *Under the null hypothesis that C has true precision p_a, the probability of materializing C is $\leq \alpha$.*

Proof. Direct from definition of p-value. The test is exact (binomial, not asymptotic). □

Theorem 6 (Family-Wise Error Rate (FWER))**.** *Let M be the total number of candidates tested. Under global null (all candidates have precision p_a),*

$$\mathbb{P}(\textit{any false positive}) \leq M\alpha \quad \textit{(Bonferroni).}$$

With $\alpha = 10^{-3}$, $M \sim 10^5$, Bonferroni gives FWER ≤ 0.1. The actual FWER is lower due to dependence among candidates.

4.3 Type II Error (False Negative / Power)

Theorem 7 (Detection Power)**.** *For a candidate with true precision $p > p_a$, coverage n, the power is*

$$\beta(n, p) = \mathbb{P}_{\text{Binom}(n,p)}(X \geq h_{crit})$$

where $h_{crit} = \min\{h : \mathbb{P}_{\text{Binom}(n,p_a)}(X \geq h) \leq \alpha\}$.

Corollary 2 (Minimum Detectable Effect)**.** *For $n = c_{\min} = 4$, $p_a \approx 0.2$, $\alpha = 10^{-3}$: $h_{crit} = 4$ (need perfect precision). For $n = 10$, $h_{crit} = 7$ (70%). The filter requires* high precision on sufficient coverage *— this is by design to avoid overfitting sparse conjunctions.*

5 Rule Recovery Guarantees

5.1 Setting

Let the true oracle contain a set of ground-truth compositional rules $\mathcal{R}^* = \{C_j^* \to a_j\}$ with arity $k_j \leq K$. Assume:

1. (Coverage) $\forall j,\ \mathrm{cov}_{a_j}(C_j^*) \geq c_{\min}$ in the explore data.
2. (Precision) $\forall j,\ \mathrm{prec}_{a_j}(C_j^*) \geq \theta + \delta$ for some $\delta > 0$.
3. (Separation) $\forall j,\ p\text{-value}(C_j^*) \leq \alpha/2$ with high probability.

Theorem 8 (Recovery of Ground-Truth Rules)**.** *Under the above assumptions, with probability $\geq 1 - \eta$ (over data sampling), every $C_j^* \in \mathcal{R}^*$ is materialized as an ACTIVE rule.*

Proof. By Assumption 1, C_j^* passes coverage pruning at all levels $\leq k_j$. By Assumption 2 and 3, it passes the binomial filter. The algorithm evaluates all candidates up to arity K, so C_j^* is considered and materialized. □

Remark 2. *Phase 3 results: 4/6 oracle rules recovered in 100% of seeds (exact match); 2/6 recovered in 50-83%. The missed rules (cold ∧ elder → obey, warm ∧ stranger ∧ spring ∧ dawn → approach) violate Assumption 1 or 2 due to priority overlap with other rules.*

6 Phase Transition in Rule Recovery

Define the *effective branching factor* $b_{\text{eff}}(k) = |S_k^{\text{supported}}|/|S_{k-1}^{\text{supported}}|$.

Proposition 1 (Phase Transition)**.** *If $b_{\text{eff}}(k) > 1$ for all $k \leq K$, the candidate set grows exponentially and induction is intractable. If $b_{\text{eff}}(k) < 1$ for some k, the search terminates early. The transition occurs at the critical coverage threshold $c_{\min}^*$ where $\mathbb{E}[b_{\text{eff}}(k)] = 1$.*

In TR-Core, $c_{\min} = 4$ places the system in the subcritical regime ($b_{\text{eff}} < 1$ for $k \geq 3$), enabling tractable discovery of arity-4 rules.

7 Comparison to Alternative Inducers

Table 1: Complexity comparison of induction methods

Method	Time	Space	Statistical Guarantee
Apriori (this work)	$\mathcal{O}(NL + L^K)$	$\mathcal{O}(N + L)$	Exact binomial test
Decision Tree (CART)	$\mathcal{O}(NL \log N)$	$\mathcal{O}(N)$	Heuristic (impurity)
Decision List (greedy)	$\mathcal{O}(NL^K)$	$\mathcal{O}(N)$	Heuristic (lift/accuracy)
k-NN	$\mathcal{O}(NL)$ (query)	$\mathcal{O}(NL)$	None (non-parametric)
Random Forest	$\mathcal{O}(TNL \log N)$	$\mathcal{O}(TN)$	Asymptotic consistency

8 Conclusion

The Apriori induction engine in TR-Core has:

- **Worst-case time** $\mathcal{O}(L^K)$ — exponential in max arity, but $K = 4$, $L \leq 25$ makes it tractable.
- **Expected time** $\mathcal{O}((1 + \beta L)^K)$ — sub-exponential due to anti-monotonicity pruning with $c_{\min} = 4$.
- **Space** $\mathcal{O}(N + L)$ — linear in data, independent of candidate space.
- **Statistical guarantee** — exact binomial test controls per-candidate Type I error at $\alpha = 10^{-3}$; high precision threshold $\theta = 0.95$ ensures rules are behaviorally meaningful.
- **Recovery guarantee** — ground-truth compositional rules are recovered if they have sufficient coverage and precision in explore data.

The combinatorial explosion is tamed by the coverage floor $c_{\min}$, which induces a phase transition to a subcritical branching process. This is the key theoretical insight: *the induction engine is tractable precisely because the environment is sparse and the coverage threshold is set above the noise floor.*